“Nav” Navaneethakannan Arumugam

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SUMMARY

Data, AI/ML, and Software engineer who ships production systems end to end: Python, database architecture, Extract-Transform-Load (ETL) and event-driven pipelines, transformer-based NLP and Large Language Model (LLM)/Retrieval-Augmented Generation (RAG) applications, and full-stack cloud deployment with CI/CD. Turns manual workflows into automated platforms adopted org-wide: 15,000+ device registers unified into a single source of truth, commissioning cut from weeks to minutes, and 100+ automated tests gating every release across energy and enterprise applications.

EXPERIENCE

Software Developer, Merit Controls — Jan 2026 – Present

Industrial control systems integration for renewable-energy assets; NJ, USA

- Established the single source of truth for vendor device data, adopted across the product team to validate SCADA, PVPC, and MPC releases, by independently architecting the OEM database: a star-schema PostgreSQL model normalizing 15,000+ Modbus/DNP3 registers from 15+ Battery Energy Storage System (BESS) and PV-inverter manufacturers into 369 canonical tags via a transaction-safe, auto-triggering ETL.
- Cut Power Plant Controller (PPC)/SCADA commissioning setup from weeks of manual engineering to minutes by building the PVPC Transformer on the OEM database, a Python, FastAPI, and React tool auto-generating AcSELerator RTAC client, controller, and driver files for 20+ devices; released v1.0 and v1.x, with database-driven v2.0 underway.
- Reduced requirements investigation and QA/QC review from hours of searching 100-page project documents to page-cited answers by architecting a fully local RAG pipeline (Tesseract OCR, all-MiniLM-L6-v2 embeddings, Chroma vector store, hybrid dense/BM25 retrieval, Qwen-1.5B Instruct), running offline at zero per-query cost.
- Kept OEM datasets current with zero manual intervention by engineering an event-driven, n8n-orchestrated microservices stack (Docker Compose, Flask API) that re-runs the ETL automatically on new device-data ingestion.
- Caught regressions before every release by building GitHub Actions CI/CD pipelines running 100+ automated unit, integration, and regression tests (pytest, vitest).
- Enabling cloud-native deployment and org-wide access by architecting and leading the ongoing migration of the OEM database and product tooling from a private Linux VM to Microsoft Azure.

AI Engineer, Copani LLC — Jan 2025 – Jan 2026

AI-powered agentic workflows for customer-service automation; FL, USA

- Automated triage of 50+ daily customer emails at 90% accuracy by building an NLP pipeline on Hugging Face Transformers that encodes inbound messages with BERT and maps them to a curated response library via semantic matching, replacing manual reading and routing.
- Protected response quality during rollout by designing the pipeline around a human-in-the-loop verification gate, where every model-selected response was reviewed before send, allowing automation to expand only as measured accuracy justified it.
- Increased email-processing efficiency 90% by integrating the pipeline with the Gmail API to classify inbound messages and auto-draft client-specific responses under strong data-security and compliance controls.
- Advanced the system from retrieval toward generation by prototyping a T5 encoder-decoder model, enabling responses to be generated rather than selected from fixed templates and extending coverage to inquiries outside the existing response library.
- Adapted a finance-focused LLaMA3 model to domain analysis on commodity hardware by applying Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA) adapters over a quantized base model served locally through Ollama, synthesizing technical, fundamental, and sentiment signals into automated investment reports that cut manual research time for internal use.

Web Developer Intern, NSIC (National Small Industries Corporation) — Jul 2022 – Aug 2022

Government small-business enterprise; Chennai, India

- Architected and launched 10+ responsive web pages (HTML, CSS) for a government small-business enterprise portal, standardizing reusable UI components across the site.
- Cut bounce rates 20% and lifted user engagement 15% by refining those UI components and improving responsive layout behavior across screen sizes.

PROJECTS

Automated Job-Application Pipeline [n8n, Hostinger VPS, Docker, Supabase, REST APIs, Google Gemini]

- Eliminated a daily manual workflow by architecting a scheduled 26-node n8n microservice pipeline integrating six external APIs (Google Drive, Google Docs, Apify, Gemini, Gmail, Supabase) that retrieves live postings, generates tailored PDFs via a Gemini-backed keyword-extraction agent rendered to LaTeX, and delivers them unattended.
- Guaranteed unattended 7 AM execution by self-hosting the Docker-containerized n8n instance on a Linux VPS rather than a local Docker Desktop environment, eliminating the dependency on a developer machine being powered on for scheduled triggers to fire.
- Enabled OAuth 2.0 authentication against Google APIs, which reject raw IP addresses as redirect targets, by mapping the VPS to a stable DuckDNS hostname that satisfies provider callback requirements and survives IP changes without re-registering credentials.
- Prevented duplicate processing and redundant third-party API spend across runs by designing a two-tier deduplication layer: in-run URL dedup plus a Supabase (PostgreSQL) persistence check issuing all lookups concurrently via `Promise.all`, backed by a job-history table keyed on the posting URL.
- Handled long-running external jobs reliably by implementing an asynchronous polling loop (submit, status-check, conditional wait-and-retry) against the Apify Actor REST API instead of blocking on fixed sleeps.
- Hardened the pipeline against upstream data-quality issues by building a defensive parsing layer in JavaScript (multi-key field fallbacks, HTML stripping and entity decoding, plus recency and content-length filters) rejecting malformed records before downstream processing.

Small Business Index Analysis Insights [Data Engineering, Apache Spark, Streamlit]

- Produced forecast-ready datasets from 5 years of raw sales data by building a PySpark ETL pipeline that imputes missing values, removes outliers, normalizes ranges, and encodes categorical features.
- Explained 90% of variance ($R^2 \approx 0.90$) forecasting the Fiserv Small Business Index across 20 NAICS sectors and 50 U.S. states by benchmarking six ML models (XGBoost winner), deployed via Streamlit for sector/state analysis.

Airbnb Price Recommendation Model [MLOps, SQLite3, Docker, DagsHub]

- Predicted Airbnb listing prices at 92% of variance ($R^2 \approx 0.92$) by processing raw data with SQLite3, engineering features from location, property type, and amenity attributes, and hyperparameter-tuning regression models.
- Deployed as a Dockerized service with managed model storage and retrieval, tracked experiments and versioned artifacts through DagsHub, served via Streamlit app for real-time price exploration and neighborhood comparison.

NYC Taxi Tipping Pattern Analysis [Python, Pandas, Data Analytics, Visualization]

- Uncovered the drivers of passenger tipping behavior by analyzing 3 million NYC taxi trip records in Pandas across passenger, trip, and socioeconomic dimensions.
- Translated raw operational data into decision-ready insight by identifying tipping trends across payment type, fare, trip duration, surcharges, and borough income levels through exploratory analysis and visualization, supporting operational planning and service optimization.

Face Recognition Check-In System [ML Engineering, OpenCV, Python]

- Built a real-time identity check-in system by combining Haar-cascade face detection with embedding-based classification (OpenCV), recognizing enrolled individuals from a live camera feed.
- Sustained reliable recognition under varied lighting by tuning embedding-distance tolerance thresholds, rejecting unknown faces rather than misclassifying them, and logging verified check-ins to Excel-based attendance records.

Library Management System [Database Design, SQL, Normalization, Full-Stack]

- Designed a Third Normal Form (3NF) relational schema for a transactional library system, modeling books, members, loans, and reservations with ACID-compliant transactions and referential-integrity constraints, implementing SQL queries and an interface covering the full borrowing lifecycle (search, checkout, return, availability).

INNOVATION & COMPETITION

Startup Boot Camp – Elysium Security Buffalo, NY

- Secured 1st place by pitching a YOLO-based real-time crime-detection system with strong market potential.

EDUCATION

M.S. in Engineering Science (Data Science), SUNY at Buffalo, NY — Jan 2024 – May 2025
B.E. in Computer Science (AI & ML), Annamalai University, India — Jul 2019 – May 2023

SKILLS

Programming Languages: Python, SQL, R, C/C++, MATLAB, PySpark, Haskell

Frameworks: TensorFlow, Scikit-learn, Keras, XGBoost, Hugging Face Transformers, BERT, T5, LLaMA3, PEFT/LoRA Fine-Tuning, RAG, LangChain, LangGraph, FastAPI, Flask, React, Streamlit, Pandas, OpenCV, Airflow, Cron

Platforms: PostgreSQL, MySQL, SQLite3, Snowflake, Supabase, Neo4j, Chroma, Spark, Databricks, Docker, Docker Compose, Linux, VPS Self-Hosting, Azure, AWS, Git, GitHub Actions, CI/CD, REST APIs, OAuth 2.0, n8n, Ollama, Jira, Power BI, DagsHub, AcSELerator RTAC, Modbus, DNP3